

A theory-first project on cognitive load, segmentation, and learning under working-memory constraints.

Background

- Cognitive load theory explains why working-memory limits matter for learning.
- Current theory is under-formalized at the level of structure, segmentation, and outcomes.
- Segmentation can ease local burden but weaken coherence needed for transfer.

Research Questions

- How should structured instructional information be modeled under working-memory constraints?
- How do segmentation and representation shape the trade-off between local relief and preserved coherence?
- Can that trade-off predict recall and transfer?

Core Idea

- Learning material is represented as interconnected information elements.
- Intrinsic cost reflects information quantity, element complexity, and coordination demand.
- Extraneous cost reflects how the material is represented or presented.
- Segmentation can lower local burden but also increase fragmentation across parts.

Main Mechanism

- Goal-relevant processing is what remains after intrinsic and extraneous costs are subtracted from limited working-memory capacity.
- Learning outcomes depend on both available processing and preserved relational structure.
- Recall should depend more on available processing; transfer should be more sensitive to fragmentation.

This is a theory-first modeling project with a proposed pilot validation study.

Hypotheses

- **H1:** Moderate segmentation will outperform no segmentation when materials are dense.
- **H2:** Excessive segmentation will hurt transfer more than recall.
- **H3:** Integrated representations will outperform split representations by lowering extraneous burden.

Validation Path / Study Design

- Use structured learning materials decomposable into elements and relations.
- Compare segmentation (none / moderate / excessive) and representation (integrated / split) conditions.
- Measure prior knowledge, recall, transfer, and an optional short load rating.
- Test whether the model's directional predictions match observed learning outcomes.